# Duc A. Hoang

*Curriculum Vitae*

This CV was updated on August 13, 2023.

## Personal Information

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|---|---|
| Full name (Vietnamese) | Hoàng Anh Đức. |
| Name (in publications) | Duc A. Hoang. |
| Name (Japanese Katakana) | ホアンアンドゥック. |
| Nationality | Vietnamese. |
| Gender | Male. |
| Languages | Vietnamese (Native) and English (Professional working proficiency). |

## Current Position

As of February 01, 2023, I am a **Lecturer** at Faculty of Mathematics, Mechanics, and Informatics, VNU University of Science (Hanoi, Vietnam).

## Contact Information


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|---|---|
| Email | hoanganhduc@hus.edu.vn |
| Email | anhduc.hoang1990@gmail.com |
| Institution | Faculty of Mathematics, Mechanics, and Informatics, VNU University of Science, 334 Nguyen Trai, Thanh Xuan, Hanoi, 100000 Vietnam. |
| Personal Webpage | https://hoanganhduc.github.io/ |
| ORCID | 0000-0002-8635-8462 |


## Research Interests

- Graph Algorithms.
- Combinatorial Reconfiguration.

## Education

Apr. 2015 - Jun. 2018 **PhD Degree in Information Science**

- **Institution:** Japan Advanced Institute of Science and Technology (Ishikawa, Japan).
- **Supervisor:** Ryuhei UEHARA.
- **Thesis Title:** Independent Set Reconfiguration and Related Problems for Some Restricted Graphs.

Apr. 2013 - Mar. 2015 **Master Degree in Information Science**

- **Institution:** Japan Advanced Institute of Science and Technology (Ishikawa, Japan).
- **Supervisor:** Ryuhei UEHARA.
- **Thesis Title:** The Independent Set Reconfiguration Problem on Some Restricted Graphs.

Sep. 2008 - Mar. 2013 **Bachelor Degree in Mathematics**

- **Institution:** VNU University of Science (Hanoi, Vietnam).
- **Thesis Advisor:** Thi Ha Duong PHAN.
- **Thesis Title:** The Matrix-Tree Theorem and Some Related Problems.
- Advanced Undergraduate Program in Mathematics.

## Employment

Feb. 01, 2023 - present **Lecturer**
Department of Informatics, Faculty of Mathematics, Mechanics, and Informatics, VNU University of Science (Hanoi, Vietnam).

Jun. 16, 2021 - Jan. 31, 2023 **Postdoctoral Researcher**
Group of Computer Algorithms, Graduate School of Informatics, Kyoto University (Kyoto, Japan).
Supervisor: Shin-ichi MINATO.
B01 Group, AFSA Project (supported by KAKENHI Grant Number 20H05964).

Apr. 01, 2021 - Jun. 15, 2021 **Research Assistant**
Kyutech Algorithms Group, School of Computer Science and Systems Engineering, Kyushu Institute of Technology (Fukuoka, Japan).
Supervisor: Toshiki SAITOH.

Apr. 01, 2019 - Mar. 31, 2021 **Postdoctoral Researcher**
Kyutech Algorithms Group, School of Computer Science and Systems Engineering, Kyushu Institute of Technology (Fukuoka, Japan).
Supervisor: Toshiki SAITOH.

Sep. 05, 2018 - Dec. 31, 2018 **Lecturer**
Department of Informatics, Faculty of Mathematics, Mechanics, and Informatics, VNU University of Science (Hanoi, Vietnam).

## Research Grants

Jul. 28, 2023 - Jul. 28, 2024 **VNU University of Science Basic-Level Grant**

- **Grant Number:** TN.23.04.
- **Project Title:** On the complexity of some reconfiguration problems on graphs under some distance constraints.
- **Role:** Principal Investigator.

Aug. 30, 2019 - Mar. 31, 2021 **JSPS KAKENHI Grant-in-Aid for Research Activity start-up**

- **Grant Number:** 19K24349.
- **Project Title:** A study on reconfiguration problems under Token Sliding and their applications.
- **Role:** Principal Investigator.

## Awards

Jun. 22, 2018 JAIST Outstanding Performance Award for doctoral students.

## Research Visits

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| Jan. 01, 2024 - Dec. 31, 2024 | Vietnam Institute for Advanced Study in Mathematics (Hanoi, Vietnam).<br>Supported by VIASM's one-year postdoctoral fellowship. |
| Apr. 07, 2021 - Apr. 08, 2021 | Group of Computer Algorithms (Minato Lab), Graduate School of Informatics, Kyoto University (Kyoto, Japan).<br>Host: Shin-ichi MINATO. |
| Dec. 23, 2019 - Dec. 25, 2019 | Faculty of Advanced Science and Technology, Kumamoto University (Kumamoto, Japan).<br>Host: Yota OTACHI. |
| Apr. 01, 2016 - Jul. 08, 2016 | Algorithm Theory Lab, Graduate School of Information Sciences, Tohoku University (Sendai, Japan).<br>Host: Xiao ZHOU and Takehiro ITO. |

## Research Activities

Research Mentor

**Vietnam Polymath REU**

- **Academic Year:** 2023-2024
- **Website:** `http://vietnampolymathreu.viasm.edu.vn/`
- **Proposed Topic:** On the structure and realizability of some reconfiguration graphs

(Sub-)Reviewer

**Journal**

- Graphs and Combinatorics (2023)
- Electronic Journal of Combinatorics (2023)
- Journal of Information Processing (2020)
- Theoretical Computer Science (2018, 2019, 2021)
- Discrete Applied Mathematics (2018)
- IEICE TRANSACTIONS on Fundamentals of Electronics, Communications and Computer Sciences (2017, 2019)

**Conference**

- ISAAC 2023
- ISAAC 2022
- WALCOM 2021, ICALP 2021, MFCS 2021
- WG 2020, COCOON 2020, ISAAC 2020
- MFCS 2019
- COCOON 2018

## Teaching Experiences

A list of courses I have participated in as a **Lecturer** or **Teaching Assistant**. The courses in English (Vietnamese) are described in English (Vietnamese).

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|---|---|
| Feb. 6, 2023 - Jun. 02, 2023 | **Lecturer** – VNU-HUS MAT2315: Phương pháp nghiên cứu khoa học (tiểu luận) |
| Feb. 6, 2023 - Jun. 02, 2023 | **Lecturer** – VNU-HUS MAT3500 3: Toán rời rạc |
| Feb. 6, 2023 - Jun. 02, 2023 | **Lecturer** – VNU-HUS MAT3500 2: Toán rời rạc |
| Sep. 10, 2018 - Dec. 13, 2018 | **Lecturer** – VNU-HUS MAT3302: Toán rời rạc (bài tập) |
| Sep. 06, 2018 - Dec. 13, 2018 | **Lecturer** – VNU-HUS MAT3302 2TNT: Toán rời rạc (bài tập) |
| Oct. 11, 2017 - Nov. 30, 2017 | **Teaching Assistant** – JAIST I216: Computational Complexity and Discrete Mathematics. |
| Apr. 12, 2017 - Jun. 02, 2017 | **Teaching Assistant** – JAIST I216: Computational Complexity and Discrete Mathematics. |
| Oct. 12, 2016 - Dec. 01, 2016 | **Teaching Assistant** – JAIST I216: Computational Complexity and Discrete Mathematics. |

Apr. 08, 2015 - Jun. 05, 2015 **Teaching Assistant** – JAIST I216: Computational Complexity and Discrete Mathematics.

## Co-authors (in alphabetical order)

David Avis, Erik D. Demaine, Martin L. Demaine, Eli Fox-Epstein, Takehiro Ito, Amanj Khorramian, Hirotaka Ono, Yota Otachi, Akira Suzuki, Ryuhei Uehara, Tsuyoshi Yagita, Takeshi Yamada.

## Publications

In all of my publications, I use the name Duc A. Hoang. A list of my publications can also be found at DBLP and Google Scholar. Some of them are available as e-prints at arXiv. All of my publications (including unpublished manuscripts) are listed at `https://hoanganhduc.github.io/publications/`.

### Journal

[4] David Avis and **Duc A. Hoang**. "On Reconfiguration Graphs of Independent Sets under Token Sliding". In: *Graphs and Combinatorics* 39.3 (2023). (article 59). DOI: `10.1007/s00373-023-02644-w`.

[3] **Duc A. Hoang**. "TS-Reconfiguration of $k$-Path Vertex Covers in Caterpillars for $k \geq 4$". In: *Theory and Applications of Graphs* 10.1 (2023). (article 8). DOI: `10.20429/tag.2023.10108`.

[2] **Duc A. Hoang**, Akira Suzuki, and Tsuyoshi Yagita. "Reconfiguring $k$-path vertex covers". In: *IEICE Transactions on Information and Systems* E105.D.7 (2022), pp. 1258–1272. DOI: `10.1587/transinf.2021EDP7177`.

[1] Erik D. Demaine, Martin L. Demaine, Eli Fox-Epstein, **Duc A. Hoang**, Takehiro Ito, Hirotaka Ono, Yota Otachi, Ryuhei Uehara, and Takeshi Yamada. "Linear-time algorithm for sliding tokens on trees". In: *Theoretical Computer Science* 600 (2015), pp. 132–142. DOI: `10.1016/j.tcs.2015.07.037`.

### Refereed International Conference

[7] **Duc A. Hoang**. "On the complexity of distance-$d$ independent set reconfiguration". In: *Proceedings of WALCOM 2023*. Ed. by Bertrand M. T. Lin, Chun-Cheng Lin, and Giuseppe Liotta. Vol. 13973. LNCS. Springer, 2023, pp. 254–266. DOI: `10.1007/978-3-031-27051-2_22`.

[6] **Duc A. Hoang**, Akira Suzuki, and Tsuyoshi Yagita. "Reconfiguring $k$-path vertex covers". In: *Proceedings of WALCOM 2020*. Ed. by M. Sohel Rahman, Kunihiko Sadakane, and Wing-Kin Sung. Vol. 12049. LNCS. Springer, 2020, pp. 133–145. DOI: `10.1007/978-3-030-39881-1_12`.

[5] **Duc A. Hoang**, Amanj Khorramian, and Ryuhei Uehara. "Shortest reconfiguration sequence for sliding tokens on spiders". In: *Proceedings of CIAC 2019*. Ed. by Pinar Heggernes. Vol. 11485. LNCS. Springer, 2019, pp. 262–273. DOI: `10.1007/978-3-030-17402-6_22`.

[4] **Duc A. Hoang**, Eli Fox-Epstein, and Ryuhei Uehara. "Sliding tokens on block graphs". In: *Proceedings of WALCOM 2017*. Ed. by Sheung-Hung Poon, Md. Saidur Rahman, and Hsu-Chun Yen. Vol. 10167. LNCS. Springer, 2017, pp. 460–471. DOI: `10.1007/978-3-319-53925-6_36`.

[3] **Duc A. Hoang** and Ryuhei Uehara. "Sliding tokens on a cactus". In: *Proceedings of ISAAC 2016*. Ed. by Seok-Hee Hong. Vol. 64. LIPIcs. Schloss Dagstuhl–Leibniz-Zentrum fuer Informatik, 2016, 37:1–37:26. DOI: `10.4230/LIPIcs.ISAAC.2016.37`.

[2] Eli Fox-Epstein, **Duc A. Hoang**, Yota Otachi, and Ryuhei Uehara. "Sliding token on bipartite permutation graphs". In: *Proceedings of ISAAC 2015*. Ed. by Khaled Elbassioni and Kazuhisa Makino. Vol. 9472. LNCS. Springer, 2015, pp. 237–247. DOI: `10.1007/978-3-662-48971-0_21`.

[1] Erik D. Demaine, Martin L. Demaine, Eli Fox-Epstein, **Duc A. Hoang**, Takehiro Ito, Hirotaka Ono, Yota Otachi, Ryuhei Uehara, and Takeshi Yamada. "Polynomial-time algorithm for sliding tokens on trees". In: *Proceedings of ISAAC 2014*. Ed. by Hee-Kap Ahn and Chan-Su Shin. Vol. 8889. LNCS. Springer, 2014, pp. 389–400. DOI: `10.1007/978-3-319-13075-0_31`.

### Thesis/Dissertation

[2] **Duc A. Hoang**. "Independent set reconfiguration and related problems for some restricted graphs". PhD thesis. Japan Advanced Institute of Science and Technology, June 2018. URL: `http://hdl.handle.net/10119/15431`.

[1] **Duc A. Hoang**. "The independent set reconfiguration problem on some restricted graphs". MA thesis. Japan Advanced Institute of Science and Technology, Mar. 2015. URL: http://hdl.handle.net/10119/12643.